

Synthesis and studies of two proton–receptor fluorescent probes based on 1,8-naphthalimide

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Two isomeric proton–receptor fluorescent probes **1** and **2** were designed and synthesised by employing the photoinduced electron transfer (PET) principle of a ‘fluorophore–spacer–receptor’ module. Their photophysical characteristics were investigated in organic solvents and Britton–Robinson buffer/EtOH (1:1, v/v) solution. Probe **1** was sensitive in the stronger acidic region of 2.21–4.56, while probe **2** displayed a good response towards protons within the pH scope of 4.10–6.09. The specificity of the probes toward protons in commonly used buffer solutions and in the presence of metal cations, Na(I), K(I), Ca(II), Mg(II), Al(III), Pb(II), Fe(III), Ni(II), Zn(II), Cu(II), Hg(II), Ag(I), Co(II), Cr(III), Mn(II), and Cd(II) ions, was studied by monitoring the changes in their fluorescence intensity. The tested metal ions exerted no noticeable influence. The results obtained show that the synthesised proton–receptor fluorescent probes have the capability to act as pH-sensitive probes for monitoring pH variations in the range 2.21–6.09.

Introduction

Within the space of a few years, the science of the recognition and sensing of environmentally and biologically significant metal ions has become an area of considerable interest to researchers working in the field of chemical sensors [1–4]. Among the many available chemical probes, fluorescence probes and switches have been widely explored owing to their unique advantages of high selectivity and high speed [5–8]. The naphthalimide derivatives constitute a unique class of environmentally sensitive fluorophores [9–11]. 4-Substituted 1,8-naphthalimide derivatives possess the attractive features of bright yellow-green fluorescence and excellent photostability qualities, which in turn enable them to enjoy a wide range of applications as DNA-cleaving molecules [12], fluorescent markers in biology [13], fluorescence sensors and copolymer [14–19], and ion probes [20,21].

Protons are one of the most important targets of interest owing to the fact that many biological and geochemical processes occurring in fresh water, sea water, and marine sediments involve strong pH changes [22]. Recent reports discussed a relation between abnormal pH values and inappropriate cell function observed in some common disease types such as cancer and Alzheimer’s [23,24]. Thus, the recognition and detection of protons have been an especially active research area. In spite of the breakthroughs recorded for the electrochemical method of pH measurement, it is not without its drawbacks, an example of which is the possibility of the signal being influenced by electromagnetic fields and aggressive analyses (alkaline solutions or F[−] ions), which can result in the destruction of the glass surface of the electrode. To counteract these obvious disadvantages, many researchers have shifted attention to the use of pH sensors owing to their attractive qualities of versatility, easiness of miniaturisation, cheapness, and possibility of remote sensing [25]. Without doubt, the use of fluorescent sensors for pH measurements is an excellent choice [26].

There are a large number of fluorescent probes for pH detection that have been well documented in the literature. Some of these probes have worked well in the neutral pH range of 6–8 [27–29], while others have displayed optimal working conditions in the weakly acidic pH range of 4–6 [30–33]. Very few probes respond to a pH value below 4 [34], and their pK_a values are below 2.5, which are suitable for extremely acidic conditions [35,36]. Hence, there is a need for the development of probes that could detect a pH value of 4 and even 3.

Accordingly, the present research paper presents the synthesis and characterisation of two new proton–receptor fluorescent probes **1** and **2**, obtained by the introduction of a piperazinyl group and an amide group. Such a design strategy enhances their capacities to combine with hydrogen ions to form hydrogen bonds, which in turn facilitates their solubility in mixed aqueous solutions, thereby rendering them capable of monitoring pH variations under acidic conditions.

Experimental

Materials

All solvents and reagents were obtained from commercial sources and used without further purification unless for special needs. Double-distilled water was used throughout all experiments. Britton–Robinson (B–R) buffer was prepared with 40 mM acetic acid, boric acid, and phosphoric acid. The pH values were suitably adjusted using dilute hydrochloric acid. Intermediate compounds *N*-allyl-4-bromo-1,8-naphthalimide, *N*-allyl-4-piperazinyl-1,8-naphthalimide, and allylpiperazine were synthesised as described previously in the literature [37–39]. 4-Bromo-1,8-naphthalic anhydride, iminodiacetic acid diethyl ester, and ammonia solution were obtained from commercial sources.

Instruments

The C, H, and N contents were determined using a Carlo Erba (Italy) 1106 elemental analyser. Melting point analyses were

carried out on an XD-4 digital micro melting point apparatus. ^1H and ^{13}C NMR spectra were obtained on Varian (USA) Mercury Plus 400 spectrometers using chloroform- d and DMSO- d_6 as solvents and tetramethylsilane (TMS) as the internal reference respectively. FTIR spectra were obtained employing a Varian Scimitar 1000 spectrometer. Electro-spray ionisation mass spectra (ESI-MS) were obtained using a Bruker (USA) micrOTOF-Q system. UV spectra were measured on a LabTech (China) BlueStar Plus UV-vis Spectrophotometer. Fluorescence spectra were measured on a Lengguang Tech (China) F97 Pro spectrofluorometer. The pH values were measured on a Beckman Coulter (USA) F360 pH meter. Silica gel 60F₂₅₄ plates (Merck KGaA, Germany) were used for thin-layer chromatography (TLC).

Syntheses

The complete synthetic routes for obtaining probes **1** and **2** are shown in Schemes 1 and 2.

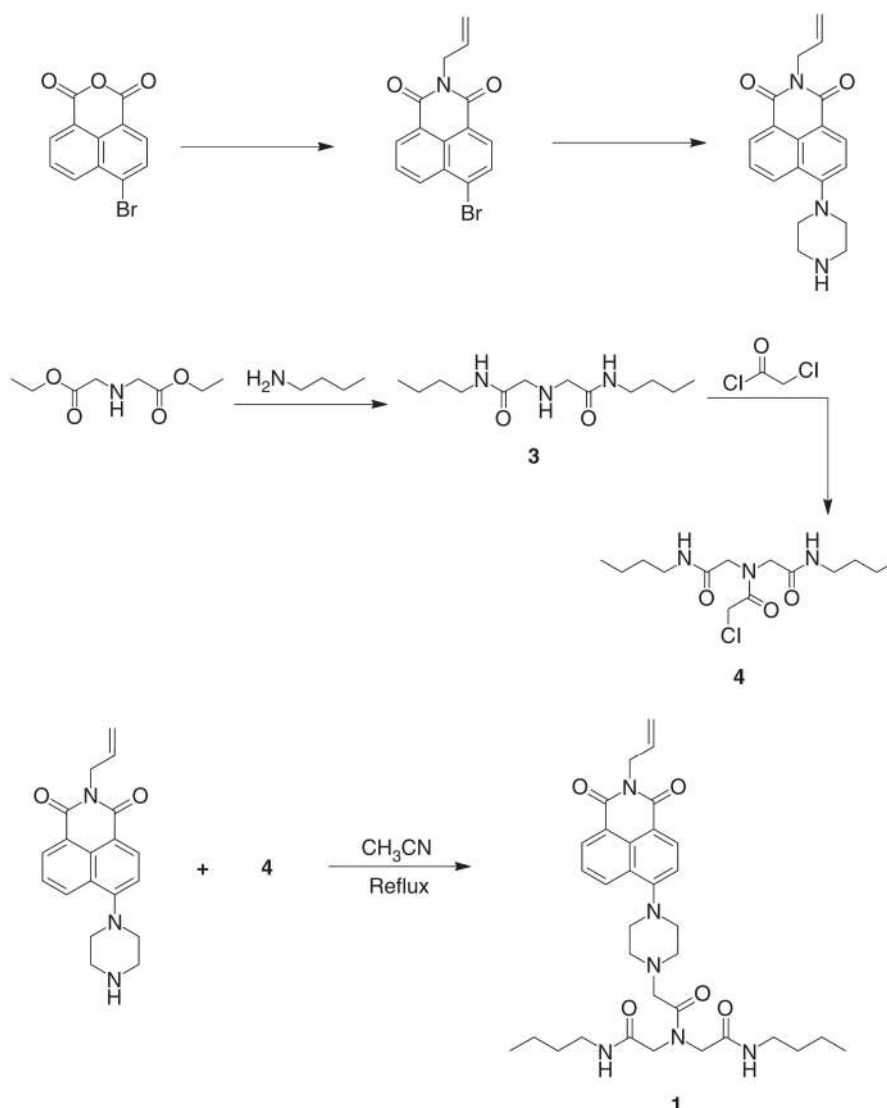
Synthesis of iminodi (N-butylacetamide) **3**

A mixture of iminodiacetic acid diethyl ester (15.1 g, 80 mmol) and butylamine (29.0 g, 397 mmol) was stirred under reflux for 15 h. Then, the reaction mixture was evaporated under reduced pressure and a white solid of

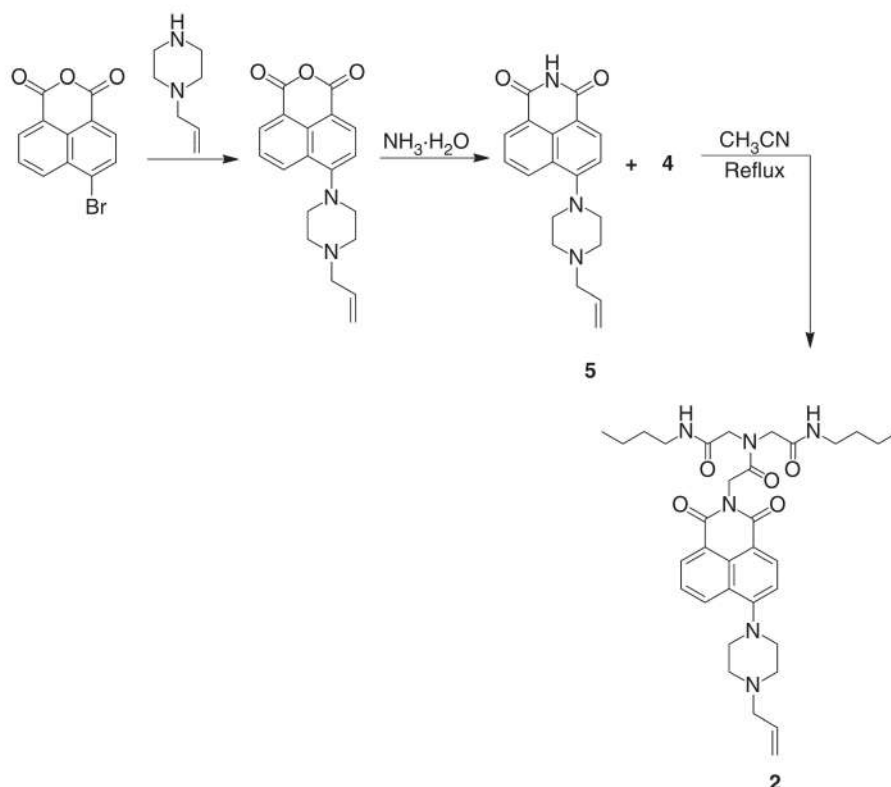
iminodi(N-butylacetamide) was formed. Yield 86%, mp 87–89 °C. FTIR (KBr, cm^{-1}) 3263, 3064, 2976, 1671. ^1H NMR (400 MHz, chloroform- d) δ 7.09 [s, 2H, (O=C–NH)₂], 3.23 (m, 4H, (–N–CH₂)₂), 3.17 [d, 4H, (O=C–CH₂)₂], 2.27 (s, 1H, –NH), 1.36 [m, 8H, (–CH₂–CH₂–)₂], 0.89 [t, 6H, –(CH₃)₂]. Elemental analysis for C₁₂H₂₅N₃O₂: calculated (%) C 59.23, H 10.35, N 17.27; found (%) C 59.32, H 10.28, N 17.21.

Synthesis of probe **1**

To a mixture of iminodi(N-butylacetamide) **3** (2.43 g, 0.01 mol), pyridine (0.79 g, 0.01 mol), and chloroform (20 ml) was added chloroacetyl chloride (1.13 g, 0.01 mol) at –10 °C to –15 °C for 10 min [40]. The reaction mixture was allowed to stir at room temperature for 5 h. After removal of the solvent under reduced pressure, 50 ml of distilled water was added, a light-yellow solid was obtained, and the solid was washed successively in parts with distilled water, 0.1 M hydrochloric acid, 0.1 M sodium hydroxide, and finally with distilled water (10 ml of each of these reagents and/or solvents was used). The solid product was filtered off and dried in vacuum to afford a crude product **4**. To a solution of *N*-allyl-4-piperazinyl-1, 8-naphthalimide (0.32 g) and acetonitrile (30 ml) was added crude product **4** (0.64 g). The resulting solution



Scheme 1 Chemical structure and synthetic route of probe **1**



Scheme 2 Chemical structure and synthetic route of probe **2**

was stirred at 82 °C for 15 h. The progress of the reaction was monitored by TLC (the TLC eluent used was a mixture of dichloromethane and ethanol, 10:1, v/v). After cooling to room temperature, the acetonitrile was evaporated at 40 °C in a rotary evaporator, followed by purification by silica gel chromatography (EtOH:CH₂Cl₂ = 1:10), which afforded the yellow-green powder probe **1**, 360 mg (0.58 mmol). Yield 56%, mp 218–220 °C. FTIR (KBr, cm⁻¹) 3430, 1691, 1660, 1590, 1380. ¹H NMR (400 MHz, chloroform-*d*) δ 9.04 (s, 1H, -NH-), 8.56 (d, 1H, *J* = 6.4 Hz, naphthalimide H-8), 8.50 (d, 1H, *J* = 8.4 Hz, naphthalimide H-6), 8.39 (d, 1H, *J* = 8.4 Hz, naphthalimide H-4), 7.67 (d, 1H, *J* = 8.0 Hz, naphthalimide H-7), 7.17 (d, 1H, *J* = 8.0 Hz, naphthalimide H-5), 6.51 (s, 1H, -NH-), 5.97 (m, 1H, -CH₂-CH=CH₂), 5.22 (q, 2H, -CH₂-CH=CH₂), 4.77 (s, 2H, -CH₂-CH=CH₂), 4.20 (s, 2H, -N-CH₂-C=O), 3.91 (s, 2H, -N-CH₂-C=O), 3.33 (s, 2H, piperazine-N-CH₂-), 3.27 [s, 8H, -N-(CH₂)₄-N-], 2.83 [s, 4H, -(NH-CH₂)₂-], 1.52 [s, 4H, -(NH-CH₂-CH₂)₂-], 1.34 [m, 4H, -(NH-CH₂-CH₂-CH₂)₂-], 0.92 [t, 6H, *J* = 7.6 Hz, -(CH₃)₂]. ¹³C NMR (DMSO-*d*₆) δ 169.9, 168.9, 168.7, 163.0, 162.4, 155.4, 132.8, 132.0, 130.4, 130.3, 128.9, 125.7, 125.1, 122.2, 116.1, 115.2, 114.7, 60.2, 52.7, 52.4, 51.7, 41.4, 38.4, 38.1, 31.1, 19.5, 19.4, 13.5. ESI-MS *m/z* 605.36 (M + H)⁺. Elemental analysis for C₃₃H₄₄N₆O₅: calculated (%) C 65.54, H 7.33, N 13.90; found (%) C 65.51, H 7.35, N 13.93.

Synthesis of probe **2**

4-Bromo-1,8-naphthalic anhydride (2.77 g, 10 mmol) and allylpiperazine were dissolved in ethylene glycol monomethyl ether (100 ml) and then heated under reflux for 10 h. The solution was allowed to cool, and the ethylene glycol monomethyl ether was distilled under vacuum to afford 4-allylpiperazinyl-1,8-naphthalic anhydride (1.9 g). 4-Allylpiperazinyl-1,8-naphthalic anhydride was added to

ammonia solution (25%, 45 ml), and the resulting solution was heated to 45 °C with stirring for 4 h. The product that precipitated on cooling was filtered, washed with 10 ml of water, and dried to give 1.2 g of crude 4-allylpiperazinyl-1,8-naphthalimide **5**. To a solution of crude product **5** (0.48 g) in acetonitrile (70 ml) was added product **4** (1.94 g). The reaction mixture was subjected to heating under reflux condition for 15 h. The reaction progress was ascertained by TLC (the TLC eluent was a mixture of dichloromethane and ethanol, 10:1, v/v). The reaction mixture was allowed to cool to room temperature, the acetonitrile was evaporated at 40 °C in a rotary evaporator, followed by purification with silica gel chromatography (EtOH:CH₂Cl₂ = 1:15) to afford the yellow-green powder probe **2**, 150 mg (0.24 mmol). Yield 16%, mp 160–162 °C. FTIR (KBr, cm⁻¹) 3430, 1660, 1590, 1380. ¹H NMR (400 MHz, DMSO-*d*₆) δ 8.93 (d, 1H, *J* = 5.2 Hz, naphthalimide H-8), 8.47 (s, 1H, -NH-), 8.45 (s, 1H, -NH-), 8.38 (d, 1H, *J* = 7.6 Hz, naphthalimide H-6), 8.21 (d, 1H, *J* = 5.6 Hz, naphthalimide H-4), 7.83 (d, 1H, *J* = 7.6 Hz, naphthalimide H-7), 7.35 (d, 1H, *J* = 8.4 Hz, naphthalimide H-5), 5.91 (m, 1H, -CH₂-CH=CH₂), 5.22 (q, 2H, -CH₂-CH=CH₂), 4.83 (s, 2H, -CH₂-CH=CH₂), 4.27 (s, 2H, -N-CH₂-C=O), 3.94 (s, 2H, -N-CH₂-C=O), 3.35 (s, 2H, -N-CH₂-), 3.10 [s, 8H, -N-(CH₂)₄-N-], 2.70 [s, 4H, -(NH-CH₂)₂-], 1.47 [s, 4H, -(NH-CH₂-CH₂)₂-], 1.38 [m, 4H, -(NH-CH₂-CH₂-CH₂)₂-], 0.91 [t, 6H, *J* = 7.2 Hz, -(CH₃)₂]. ¹³C NMR (DMSO-*d*₆) δ 170.5, 168.3, 168.2, 167.6, 163.1, 162.5, 155.8, 135.2, 132.3, 130.7, 129.1, 125.9, 125.2, 122.2, 117.6, 115.0, 60.7, 52.6, 52.4, 52.2, 52.1, 51.7, 40.6, 38.4, 38.2, 37.8, 35.7, 31.3, 31.0, 28.9, 19.5, 19.4, 19.3, 13.6, 13.5. ESI-MS *m/z* 605.36 (M + H)⁺. Elemental analysis for C₃₃H₄₄N₆O₅: calculated (%) C 65.54, H 7.33, N 13.90; found (%) C 65.50, H 7.32, N 13.95.

Optical absorption spectroscopy

Absorption spectra were obtained on a LabTech BlueStar Plus UV-vis spectrophotometer (controlled by a Pentium PC running the manufacturer-supplied software package) with a quartz cuvette size of 1×1 cm.

Fluorescence spectroscopy

Fluorescence spectra were recorded from 420 to 700 nm on a Lengguang Tech F97 Pro spectrophotometer with an excitation wavelength of 395 nm; the excitation and emission slit widths were both 5 nm. The excitation source was a 150 W Xenon lamp. Fluorescence measurement was carried out in right-angle sample geometry. A 1×1 cm quartz cuvette was used for the spectroscopic analysis. All measurements were carried out at least three times.

Tolerance of probes to H^+ over other metal ions

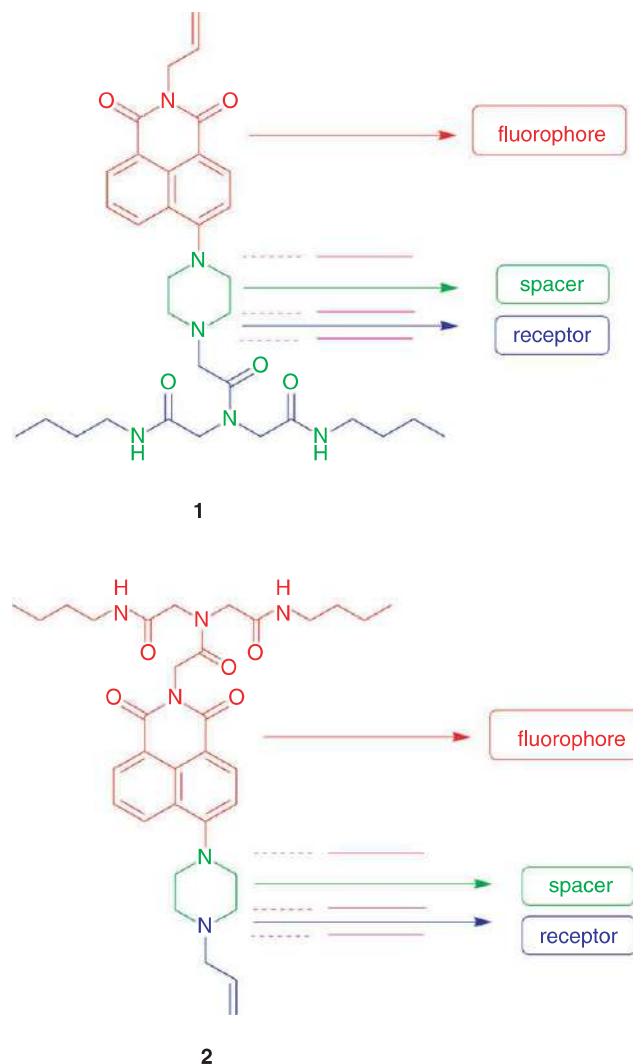
The effects of the metal cations were examined by adding 50 μ l of the metal stock solutions (1×10^{-3} M) to 2 ml of the B-R (pH 2.87/4.56 for probes **1** and **2** respectively)/EtOH (1:1, v/v) fluorophore solution (5×10^{-6} M).

Results and Discussion

Design and synthesis of probes based on the PET process

As a matter of design, our approach to obtaining pH probes **1** and **2** exploited the photoinduced electron transfer (PET) principle of a 'fluorophore-spacer-receptor' array. A fluorophore has an intrinsic signalling property, which could be easily modulated upon interaction with the recognition receptor. The spacer simply keeps the fluorophore and receptor at a short distance apart to allow ease of intramolecular PET. We employed 1,8-naphthalimide as the signalling unit so as to facilitate a weak repulsion field at the imide group and allow ease of the PET process [41]. The piperazinyl group serves as the spacer and the second nitrogen atom on the terminal substituted piperazine in the C-4 position is the analyte receptor. We employed an inhibited PET process of electron transfer from the receptor to the fluorophore. When an analytical species interacts with a receptor moiety, the photophysical properties of a fluorophore moiety undergo some changes that lead to a signal, thereby confirming guest binding [42,43]. Herein, annihilation of the PET process occurred and recovery of fluorescence of the system ensued [41,44–47]. The protonation of the piperazine *N*-amine would increase the oxidation potential of the receptor and as such would be thermodynamically infeasible for electron transfer [48,49]. Consequently, the emission intensity would be greatly increased. Thus, we expect the fluorescence to be strong in acidic media [50]. To the best of our knowledge, these are the first successful cases of chemosensors constructed from an integration of these three specific components. Scheme 3 summarises the design module we employed in obtaining pH probes **1** and **2**.

The fluorescent probes **1** and **2** were synthesised as shown in Schemes 1 and 2. They are stable in air and soluble in solvents such as DMF, DMSO, THF, chloroform, acetonitrile, and ethanol, but insoluble in Et₂O and water. The structures and purities of the target products were confirmed by conventional techniques – elemental analysis data, UV-vis, fluorescence, FTIR, MS, and ¹H and ¹³C NMR spectroscopy.



Scheme 3 Design of probes **1** and **2** based on the 'fluorophore-spacer-receptor' array [Colour figure can be viewed at wileyonlinelibrary.com]

Photophysical characteristics of probes

It is well known that the polarisation of chromophoric systems of 1,8-naphthalimide derivatives has a large influence on their photophysical properties. The photophysical properties of the two probes were investigated in B-R/EtOH (1:1, v/v) and organic solution. Table 1 presents the basic photophysical characteristics of probes **1** and **2** in alkaline and acidic conditions: absorption (λ_A) and fluorescence (λ_F) maxima, Stokes shifts, and molar extinction coefficients (ϵ). The emission of probe **1** was observed in the visible region with well-pronounced maxima (λ_F) at 516–535 nm, while the emission of probe **2** has significant maxima positioned at 519–535 nm.

Table 1 Photophysical characteristics of probes **1** and **2**

	pH	λ_A , nm	λ_F , nm	$\nu_A-\nu_F$, cm^{-1}	ϵ , $\text{dm}^3 \text{ mol}^{-1} \text{ cm}^{-1}$
Probe 1	11.58	409	535	5758	16 700
	1.81	386	516	6527	16 900
Probe 2	11.58	409	535	5758	8700
	1.81	387	519	6572	8700

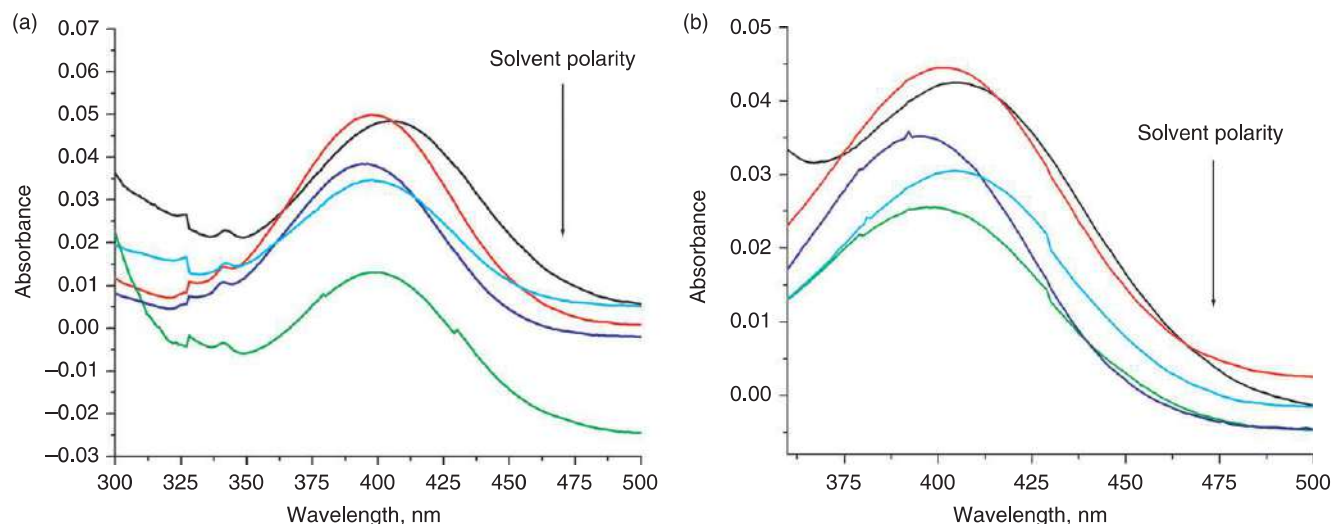


Figure 1 Absorption spectra of probe **1** (a) and probe **2** (b) in different solvents [Colour figure can be viewed at wileyonlinelibrary.com]

Figure 1 gives the absorption spectra of the two probes in different organic solvents. For both probes it was observed that the values of absorbance showed good linearity with increasing solvent polarity. Tables 2 and 3 give the respective absorption/excitation values and molar extinction coefficients of the two probes. It is clear that the molar extinction coefficient (ϵ) depends on the solvent media, and it became larger with increasing solvent polarity.

The Stokes shift ($\nu_A - \nu_F$), as a vital photophysical parameter of fluorescent compounds, gives a reflection of the discrepancies in the properties and structure of the fluorophores between the ground state S_0 and the first excited state S_1 . The Stokes shifts (cm^{-1}) were calculated by the equation [14].

$$(\nu_A - \nu_F) = \left(\frac{1}{\lambda_A} - \frac{1}{\lambda_F} \right) \times 10^7 \quad (1)$$

and the results are set out in Table 4.

Table 4 shows that the Stokes shift depends on the media and is larger in the case of polar solvents when the hydrogen bond formation or dipole–dipole interactions are favoured in comparison with non-polar media [51].

The ability of the molecules to emit absorbed light energy is characterised quantitatively by the fluorescence quantum yield (Φ_F). The quantum yield of fluorescence was also calculated using quinine sulfate ($\Phi_F = 0.54$ in $0.1 \text{ M H}_2\text{SO}_4$) as the standard according to the equation [52].

Table 2 Photophysical characteristics of probe **1** in different solvents

	DMSO	Acetonitrile	Ethanol	THF	Dichloromethane
λ_A , nm	405	398	399	396	398
λ_F , nm	532	524	529	506	507
ϵ , $\text{dm}^3 \text{ mol}^{-1} \text{ cm}^{-1}$	12 100	12 500	3300	9600	8700

Table 3 Photophysical characteristics of probe **2** in different solvents

	DMSO	Acetonitrile	Ethanol	THF	Dichloromethane
λ_A , nm	404	401	400	392	404
λ_F , nm	528	524	525	510	509
ϵ , $\text{dm}^3 \text{ mol}^{-1} \text{ cm}^{-1}$	8800	9400	5100	8000	5900

Table 4 Stokes shifts ($\nu_A - \nu_F$) of compounds in different solvents

	DMSO, cm^{-1}	Acetonitrile, cm^{-1}	Ethanol, cm^{-1}	THF, cm^{-1}	Dichloromethane, cm^{-1}
Probe 1	5955	6042	6159	5489	5402
Probe 2	5813	5854	5952	5902	5106

Table 5 Fluorescence quantum yields of probes in different solvents

	DMSO	Acetonitrile	Ethanol	THF	Dichloromethane
Probe 1	0.01	0.05	0.04	0.11	0.14
Probe 2	0.01	0.01	0.01	0.04	0.04

$$\Phi_F = \Phi_{\text{ref}} \left(\frac{S_{\text{sample}}}{S_{\text{ref}}} \right) \left(\frac{A_{\text{ref}}}{A_{\text{sample}}} \right) \left(\frac{n_{\text{sample}}^2}{n_{\text{ref}}^2} \right) \quad (2)$$

where A_{ref} , S_{ref} , n_{ref} and A_{sample} , S_{sample} , n_{sample} represent the absorbance at the excitation wavelength, the integrated emission band area, and the solvent refractive index of the standard and the sample respectively. The Φ_F values of both probes in different solvents are listed in Table 5. It is clear that the values of Φ_F for both probes tend to become higher with decreasing polarity of the solvents; this can be explained by the photoinduced electron transfer processes, which are accelerated in polar solvents [53].

pH effect on the absorption and fluorescence properties of the probes

The UV-vis spectrum of both probes were recorded in an alkaline medium at pH 11.58, and absorption bands between 300 and 500 nm with a λ_A maximum at 409 nm were observed (Fig. 2). Upon acidification to pH 1.81, the bands were blue shifted with almost no changes in their maximum intensities. As supported by the findings of Gunnlaugsson and coworkers [54], the reason for the blue shift observed is twofold: the protonation of the amine receptor, which exerted some weak charge repulsion on the 4-amino moiety of the fluorophores, and the push-pull character of the excited state in acidic conditions, which was partially reduced by the protonation of the 4-amino moiety itself [17].

The results of studying the spectroscopic properties of the sensors in the pH range 1.81–11.58 in B-R/EtOH (1:1, v/v) are presented in Fig. 3. As expected, in alkaline media the compounds showed a very weak fluorescence intensity in the range 420–700 nm, with a maximum at 535 nm (1,8-naphthalimide emission). However, upon acidification, the

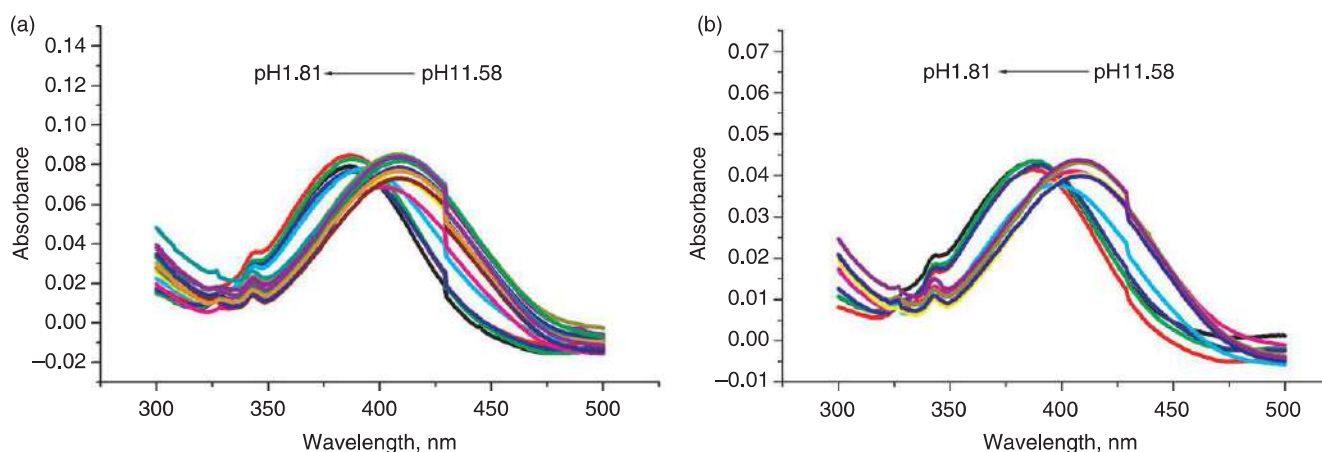


Figure 2 Absorption spectra of probe 1 (a) and probe 2 (b) in B-R/EtOH (1:1, v/v) solution at different pH values [Colour figure can be viewed at wileyonlinelibrary.com]

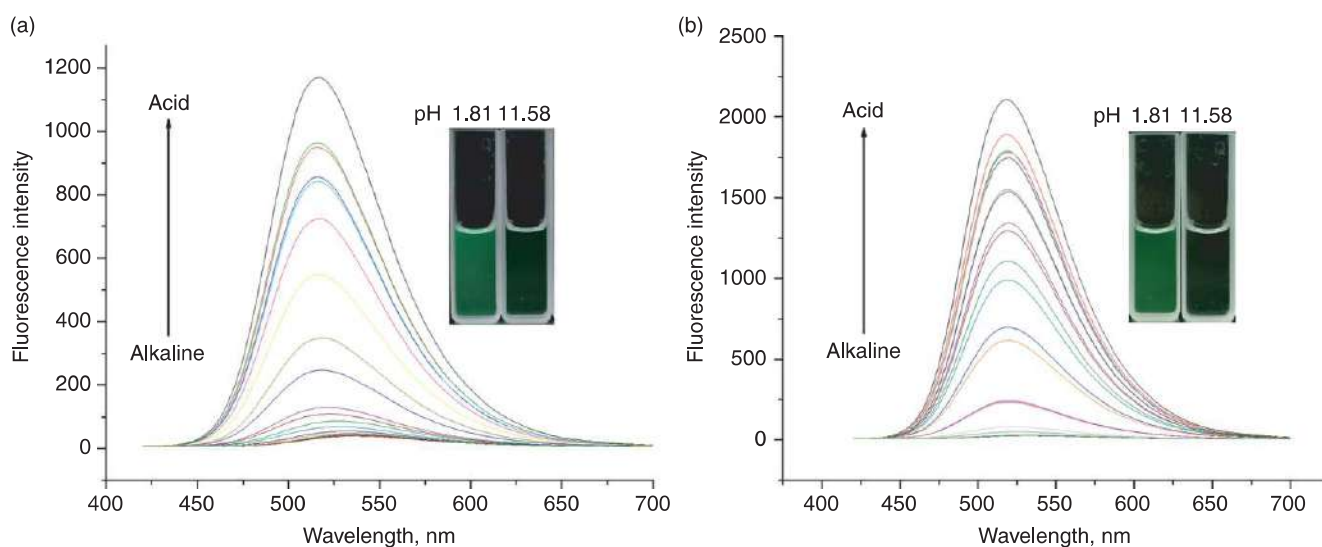


Figure 3 (a) Changes in the fluorescence spectra of probe 1 (5 μM) as a function of pH in B-R/EtOH (1:1, v/v). The pH range was from 11.58 to 1.81. (b) Changes in the fluorescence spectra of probe 2 (5 μM) as a function of pH in B-R/EtOH (1:1, v/v). The pH range was from 11.58 to 1.81. Inset: the change in colour with pH at 1.81 and 11.58 [Colour figure can be viewed at wileyonlinelibrary.com]

emission signal was gradually increased when the pH values changed from 11.58 to 1.81, and the fluorescence enhancement (FE) was 24.53-fold for probe **1** and 78.44-fold for probe **2**. Here, the FE was the ratio of maximum fluorescence intensity at pH 11.58 to that at pH 1.81. We also found that the maximum emission wavelength blue shifted from 535 to 516 nm and from 535 to 519 nm for probes **1** and **2** respectively, indicating that structural changes occurred for both probes, which can be attributed to the acidification. Moreover, taking the part of Fig. 4 located between pH = 2.21–4.56 and pH = 4.10–6.09, the pH dependence of fluorescence intensity was analysed using the equation [17].

$$\text{pH} = \text{pK}_a - \log \left[\frac{I_x - I_a}{I_b - I_x} \right] \quad (3)$$

with pK_a values of 2.75 and 2.39 being found. Based on an equation in the literature [55], we know that the calculated pK_a values imply that the probes are very sensitive around these pH values.

Furthermore, for probe **1**, the fluorescence intensity and pH value exhibited good linearity in the pH range 2.21–4.56, with a regression coefficient $R^2 = 0.9942$. In the same vein, for probe **2**, linearity between the fluorescence intensity and the pH value was observed in the pH range 4.10–6.09, with $R^2 = 0.9820$. The results indicate that the two probes are very valuable for the detection of strong acidic environments and could be combined to monitor a wider pH scale from 2.21 to 6.09.

The proposed mechanism of detecting pH in the strong acidic region may be ascribed to blockage of electron transfer from the piperazine ring to the naphthalimide subunit, thereby resulting in fluorescence enhancement. We prepared the control probes **1** and **2** and their protonated forms **1**·HCl and **2**·HCl to study the proton-binding behaviour by ^1H NMR spectroscopy. As shown in Fig. 5, upon protonation, the two allyl protons of probe **2** displayed noticeable downfield shifts of 0.21 and 0.38 ppm respectively, and other significant changes involved new signals appearing downfield at 10.69 and 11.79 ppm for

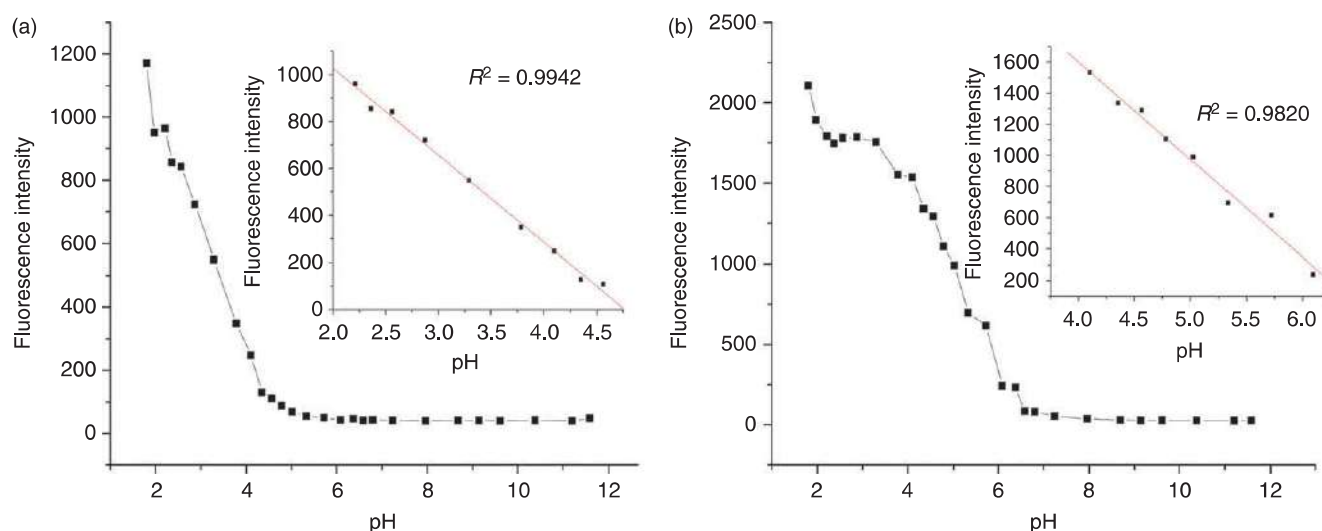


Figure 4 (a) Effect of pH on the fluorescence intensity ($\lambda_{\text{ex}} = 395$ nm) of probe **1** (5 μM) in B-R/EtOH (1:1, v/v). (b) Effect of pH on the fluorescence intensity ($\lambda_{\text{ex}} = 395$ nm) of probe **2** (5 μM) in B-R/EtOH (1:1, v/v). The insets show the linear relationship of fluorescence intensity at 517 nm and pH values from 2.21 to 4.56 ($R^2 = 0.9942$) and from 4.10 to 6.09 ($R^2 = 0.9820$) for probe **1** and probe **2** respectively [Colour figure can be viewed at wileyonlinelibrary.com]

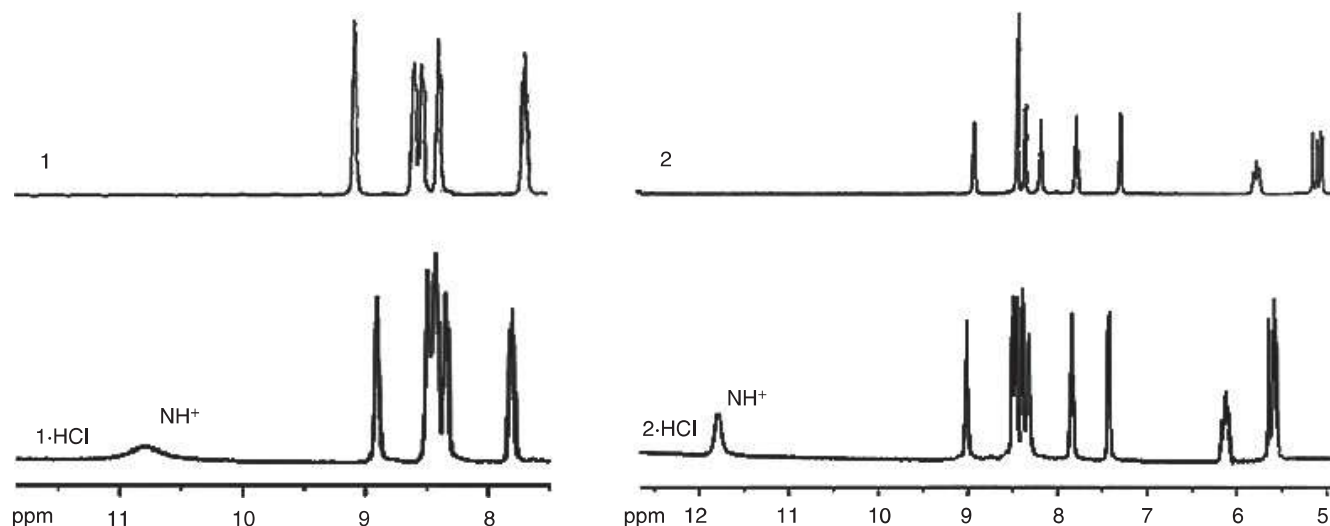
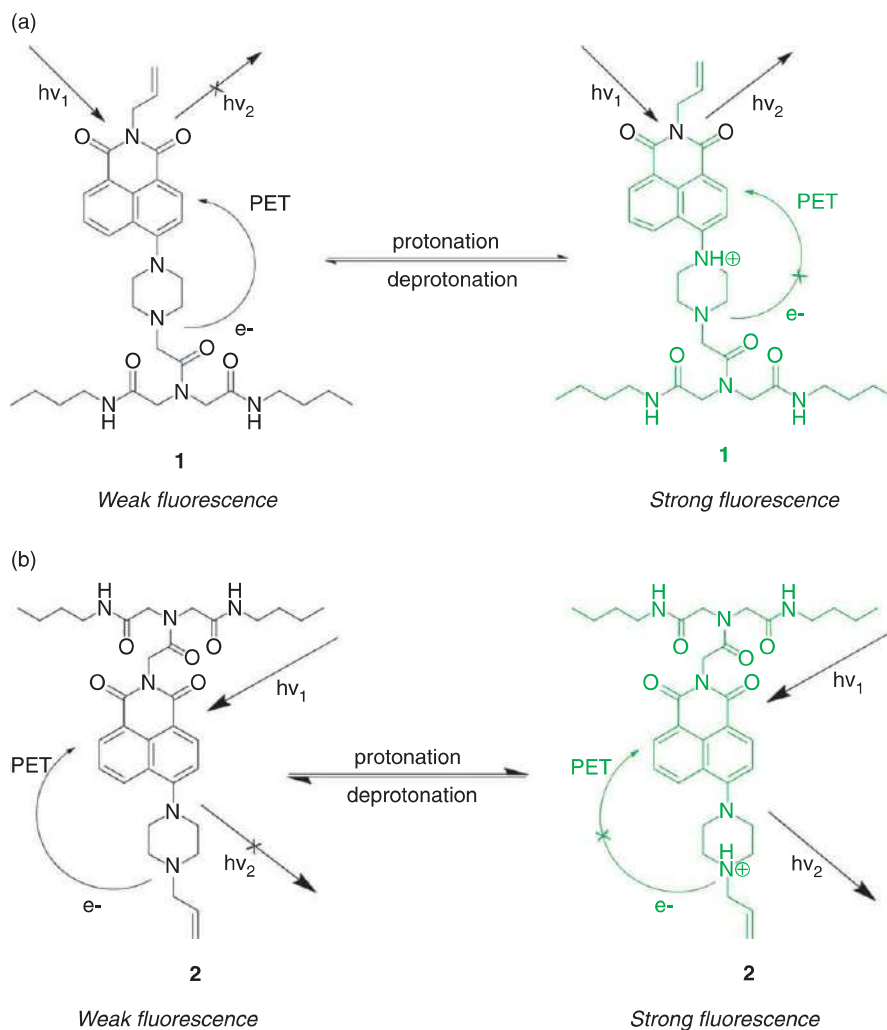


Figure 5 Partial ^1H NMR spectra of probe **1** and **1**·HCl and of probe **2** and **2**·HCl

probe **1** and **2** respectively. Therefore, the results revealed that protonation occurs at the piperazinic site and is in good agreement with similar studies in the literature [10,56]. The

proposed specific pH sensing mechanism for both probes can be described as the protonation-induced inhibition of the PET process from the second nitrogen atom on the



Scheme 4 (a) Proposed inference of pH sensing for (a) probe **1** and (b) probe **2** [Colour figure can be viewed at [wileyonlinelibrary.com](#)]

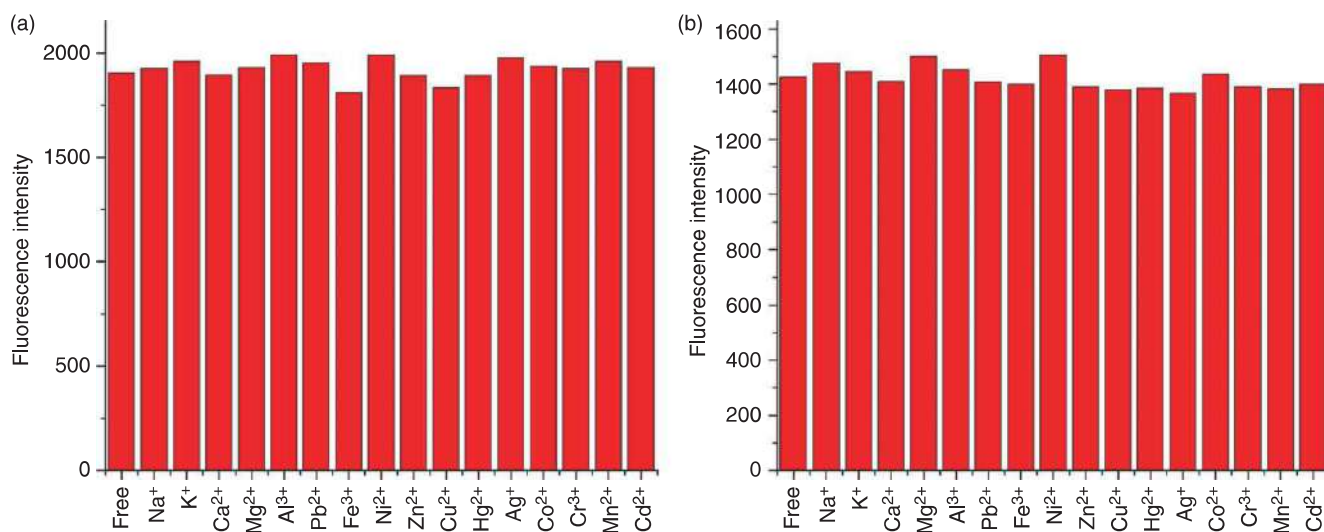


Figure 6 (a) Fluorescence responses at $\lambda_{em} = 517$ nm of probe **1** (5 μM) to diverse metal ions in Britton–Robinson (pH 2.87)/EtOH (1:1, v/v) solution. The concentrations of the metal cations are 25 μM, and the measuring temperature is 25 °C. (b) Fluorescence responses at $\lambda_{em} = 519$ nm of 5 μM probe **2** to diverse metal ions in Britton–Robinson (pH 4.56)/EtOH (1:1, v/v) solution. The concentrations of the metal cations are 25 μM, and the measuring temperature is 25 °C [Colour figure can be viewed at [wileyonlinelibrary.com](#)]

terminal substituted piperazine in the C-4 position to the naphthalimide fluorophore, thereby leading to fluorescence enhancement (Schemes 4a and b).

Influence of metal cations on the fluorescence intensity of the probes

For further evaluation of whether the pH-responsive properties can be interfered with by other metal cations, the signalling properties of probes in the presence of the other competitive metal ions [Na(I), K(I), Ca(II), Mg(II), Al(III), Pb(II), Fe(III), Ni(II), Zn(II), Cu(II), Hg(II), Ag(I), Co(II), Cr(III), Mn(II), and Cd(II) ions] were investigated as shown in Fig. 6. We prepared a series of 5 μ M concentrations of probes **1** and **2** dissolved in B–R (pH 2.87)/EtOH (1:1, v/v) and B–R (pH 4.56)/EtOH (1:1, v/v) solutions respectively. The free group containing only 5 μ M of probe displayed strong fluorescence in acidic condition. There were almost no changes in the fluorescence intensity when different metal ions were added to the acid mixed solutions, which indicates that the metal ions have no effect on the strong fluorescence caused by H⁺. Hence, we can assert that metal ions cause insignificant interference, and the two probes only respond to H⁺.

Conclusion

This article reports the synthesis of two proton-receptor PET fluorescent probes based on 1,8-naphthalimide (the fluorophore) by the introduction of a piperazinyl group (spacer) and an amide group (receptor) to enhance their abilities to combine with hydrogen ions and form hydrogen bonds. The two probes displayed intense fluorescence responses towards protons in acidic solution. The experimental results showed that these compounds could serve as novel fluorescent pH probes for further application. Meanwhile, both probes contain an unsaturated allyl group, meaning that they have the potential to be polymerised with other polymer monomers and to be more conveniently used as polymeric devices to detect pH as in strongly acidic wastewater. As time and experimental conditions are limited at the present stage, the polymerisation and practical application will be performed in follow-up studies.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Fig. S1 ¹H-NMR of iminodi(*N*-butylacetamide) **3**

Fig. S2 ¹H-NMR of probe **1**

Fig. S3 ¹H-NMR of probe **2**

Fig. S4 ¹³C-NMR of probe **1**

Fig. S5 ¹³C-NMR of probe **2**

Fig. S6 Mass spectra of probe **1**

Fig. S7 Mass spectra of probe **2**